

Bachelorarbeit

Minimale Neuberechnung Abhängigkeits-Graph basierter Regeln zur prozeduralen Welten-Generation

Maarten Behn

CGVR Universität Bremen

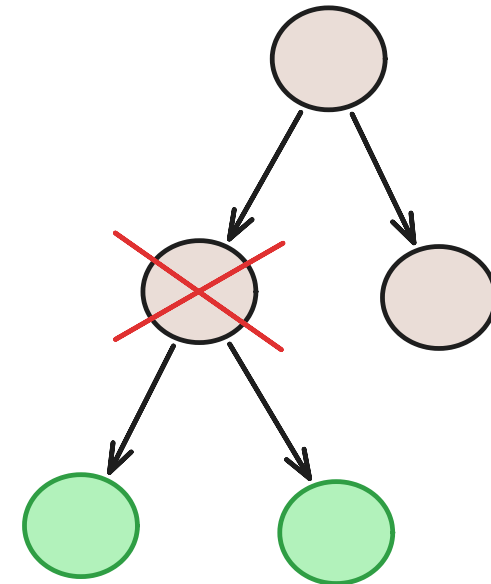
8.9.2026

Fragestellung

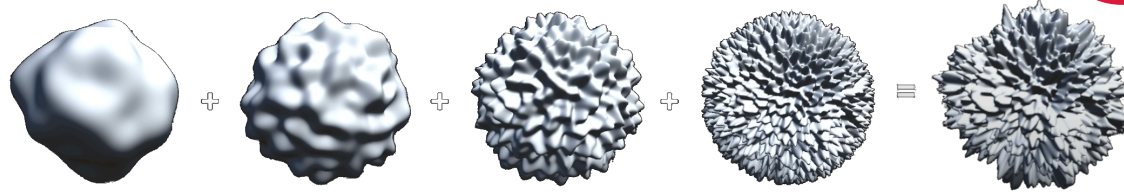
Generierte Welt



Generationsschritte

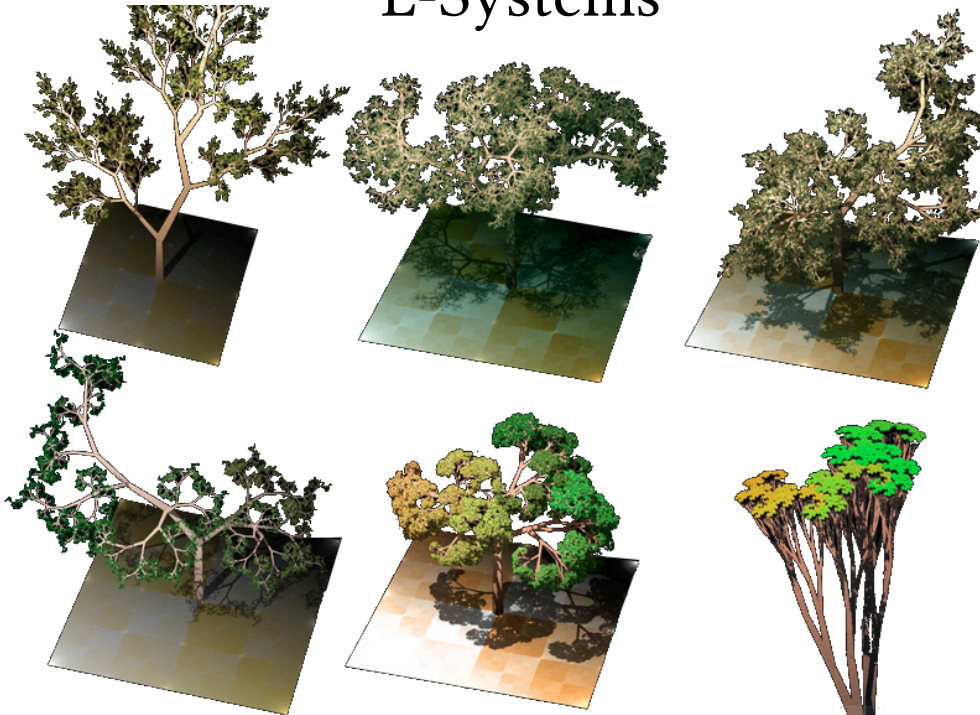


Related Work

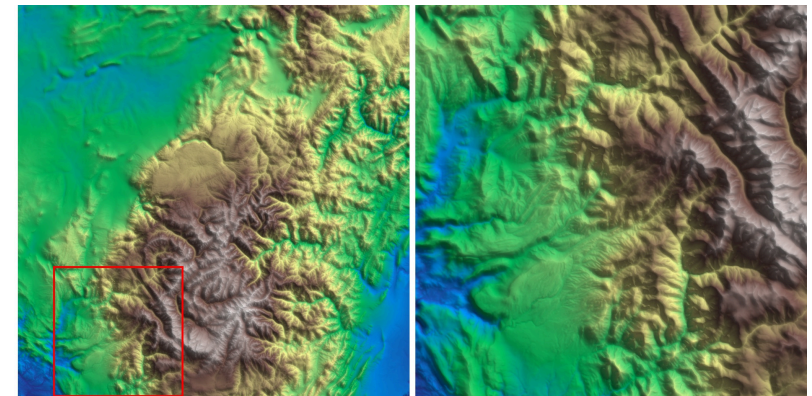


Layered Noise

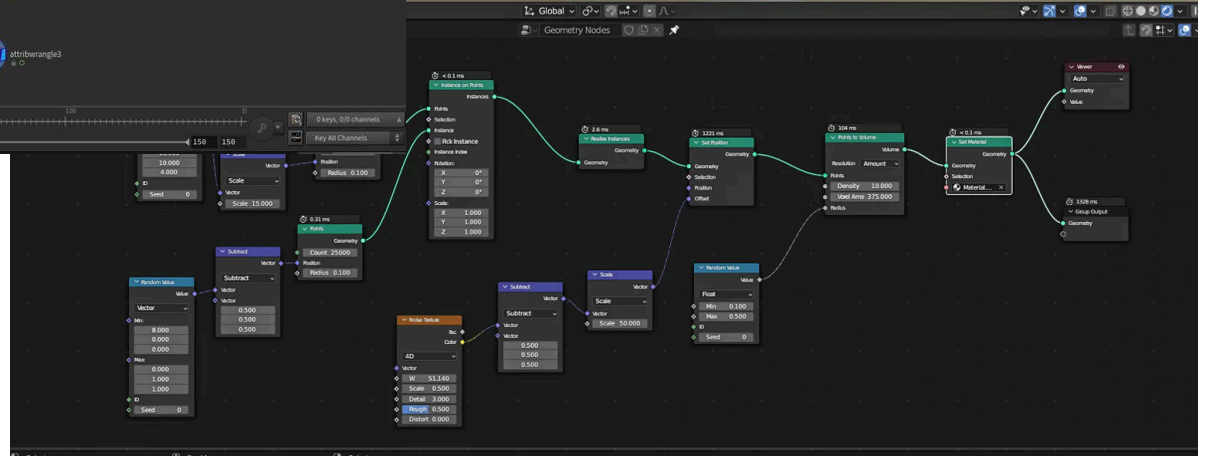
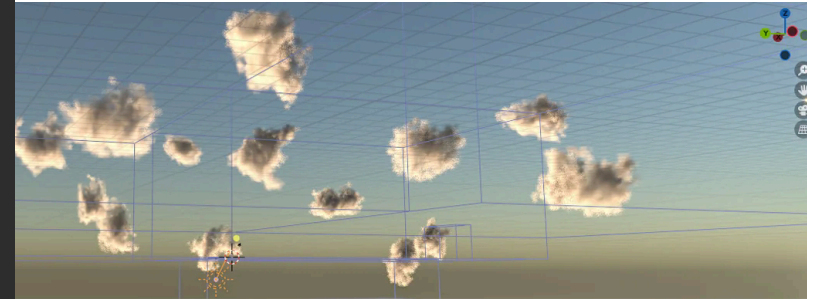
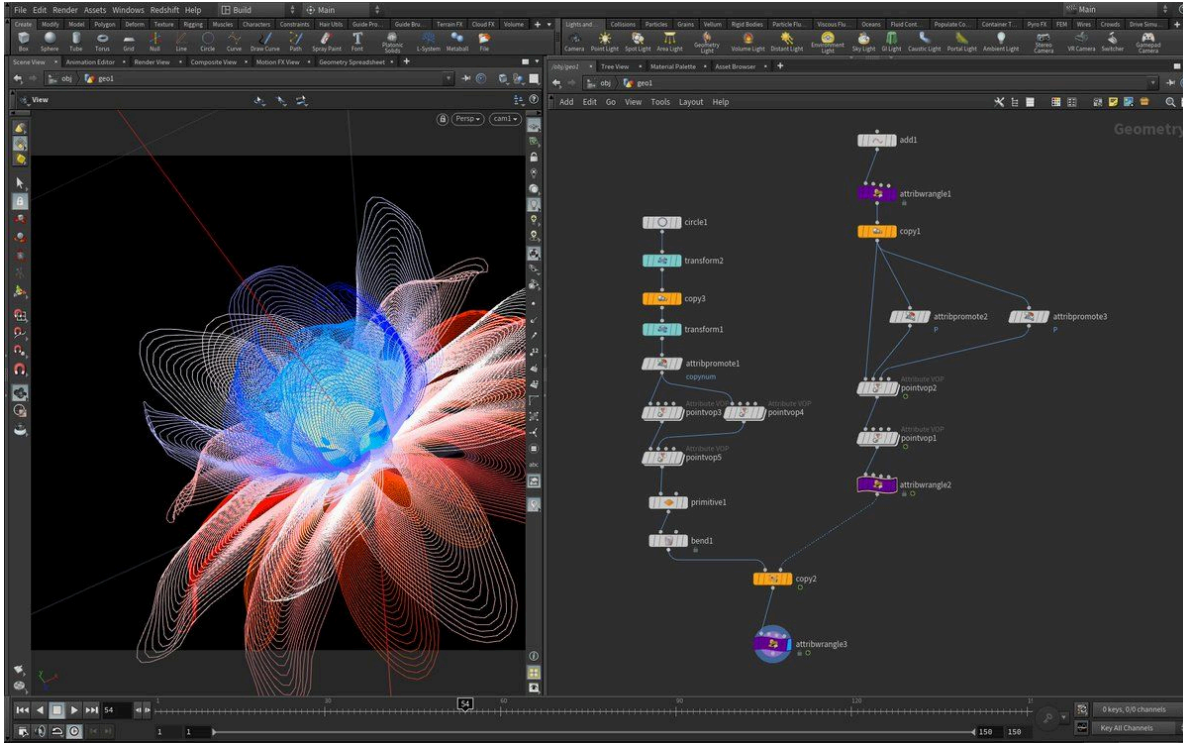
L-Systems



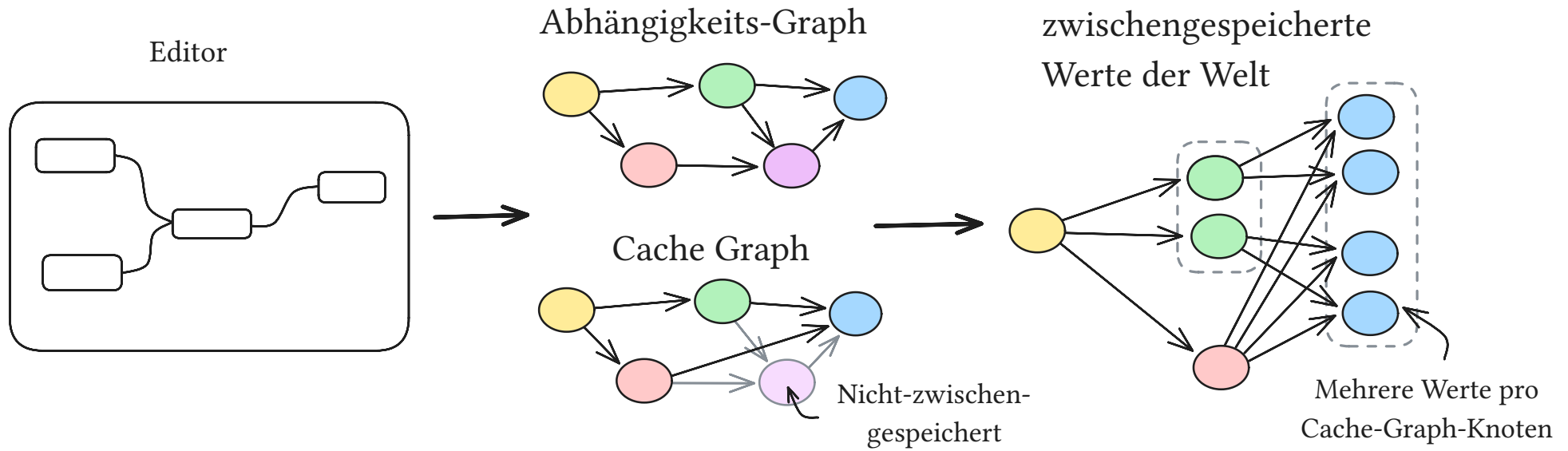
Terrain Diffusion

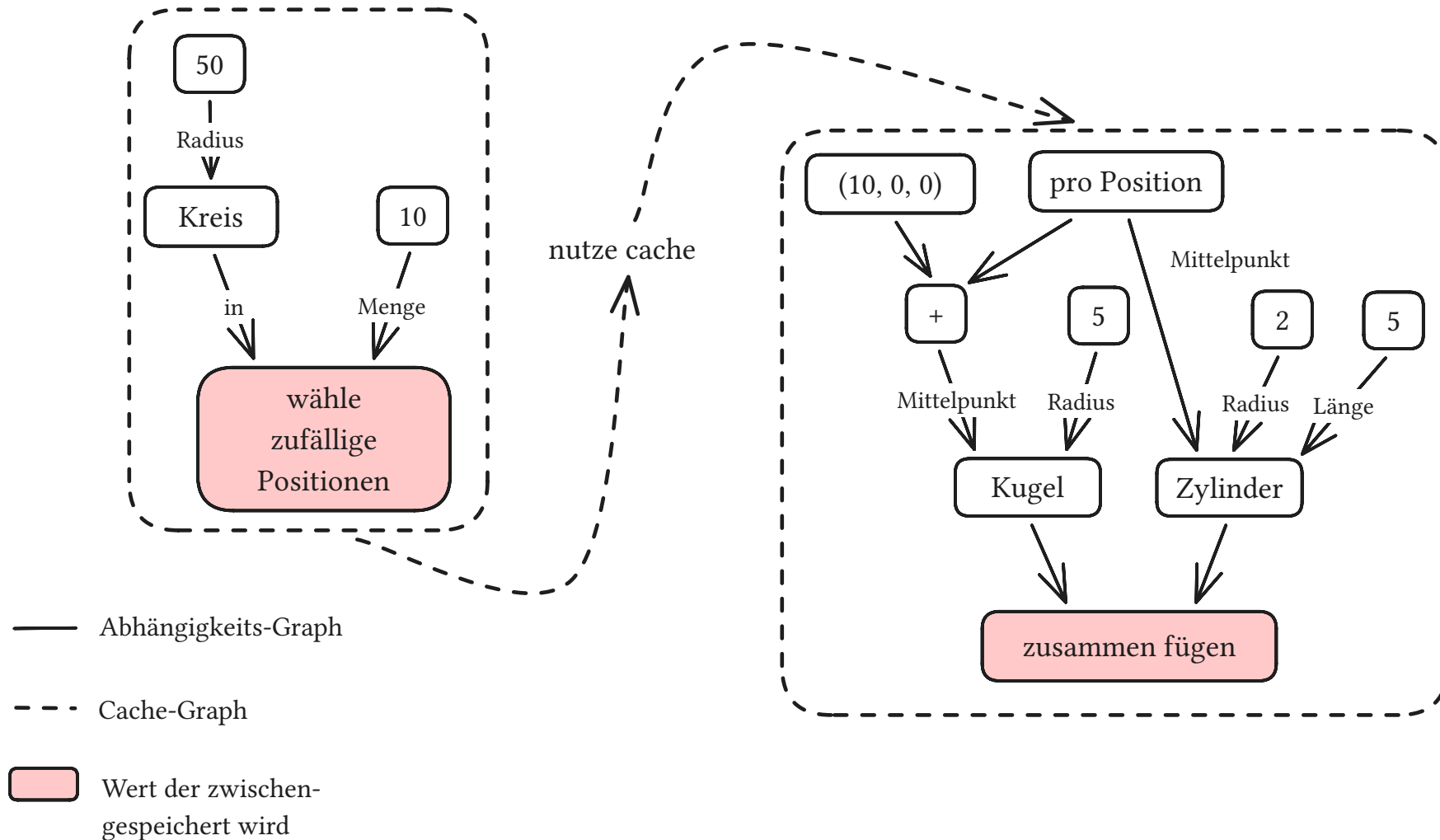


Houdini & Blender



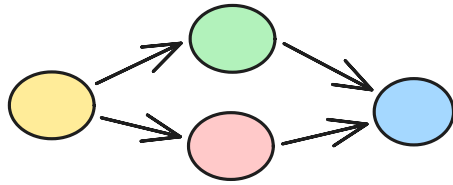
Mein Lösungsansatz



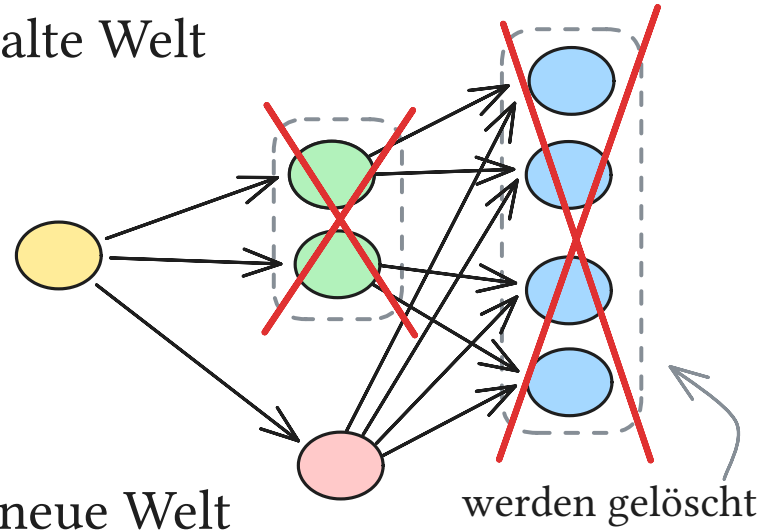


Templates vergleichen

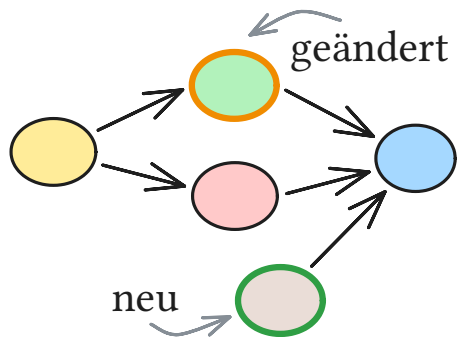
altes Template



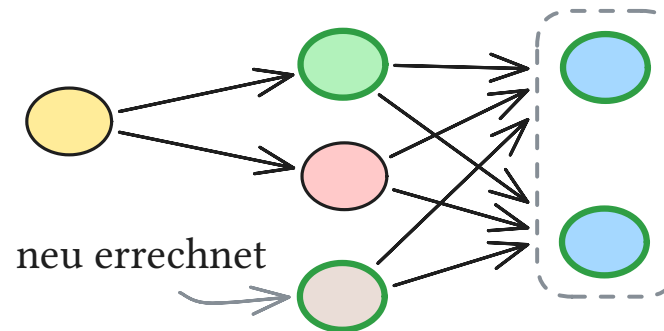
alte Welt



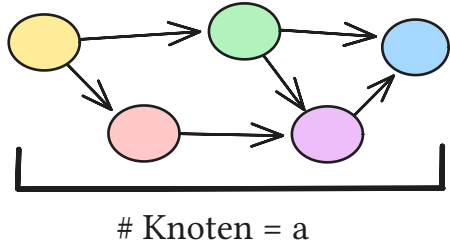
neues Template



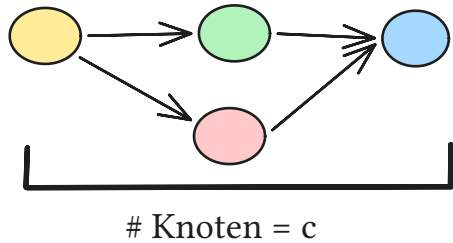
neue Welt



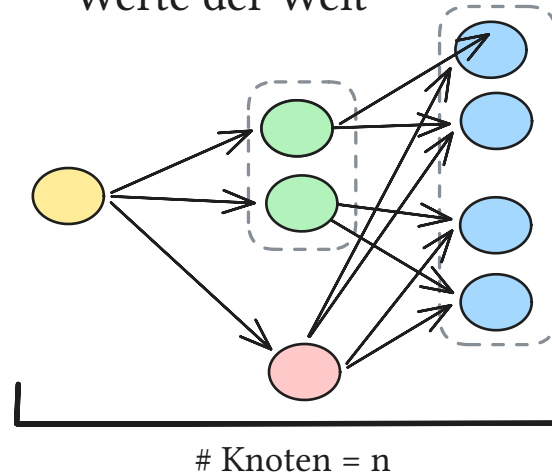
Abhängigkeits-Graph



Cache Graph



zwischengespeicherte
Werte der Welt



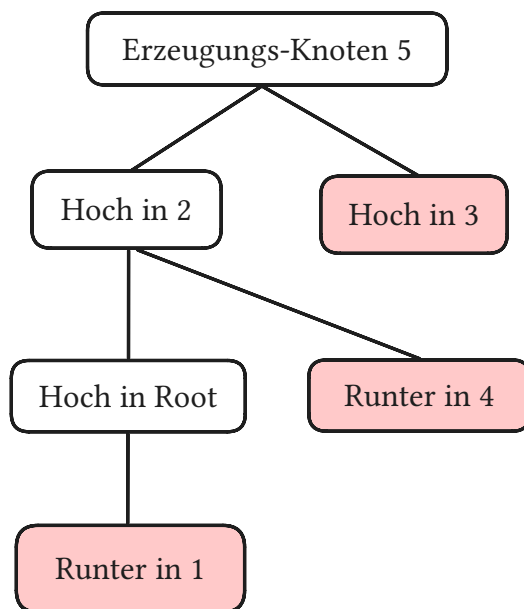
Cache-Fraktor: $c_f := \frac{a}{c}$

Branch-Fraktor: b_f

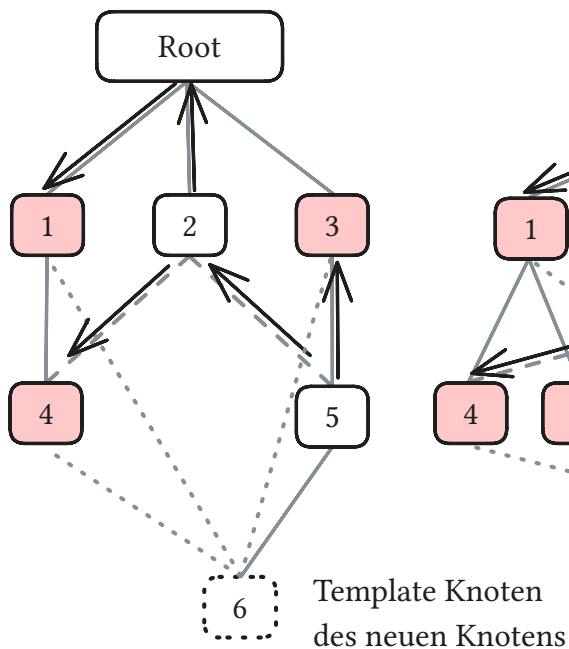
Neuberechnungsfaktor: g_f

$$O(n) = O\left((g_f c_f a)^{b_f}\right) = O(a^{b_f})$$

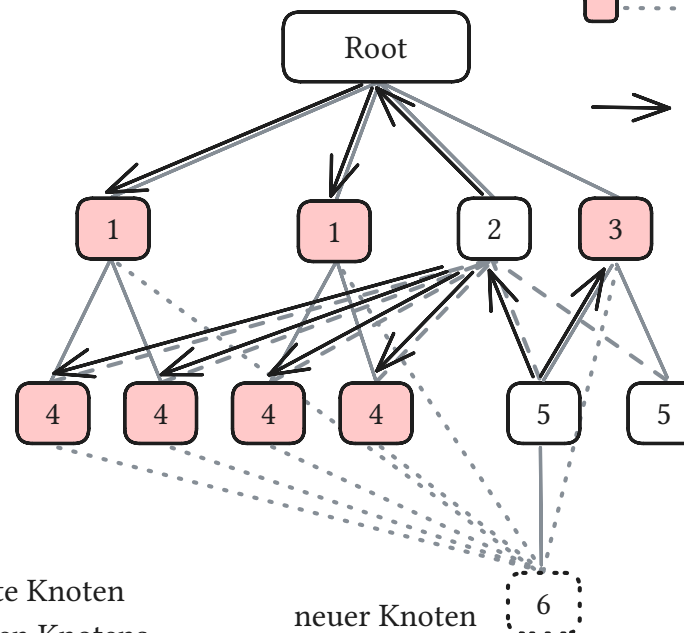
Baum an relativen Schritten



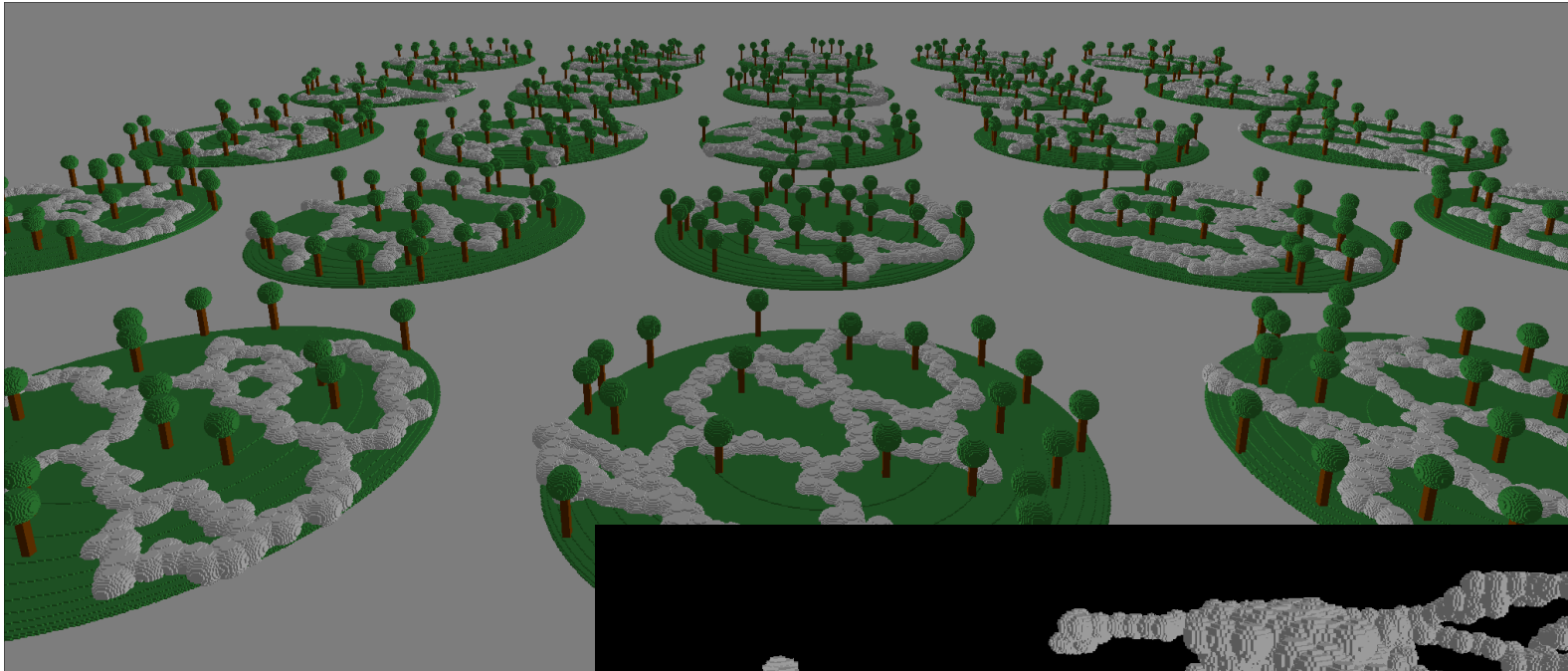
Template



Generator

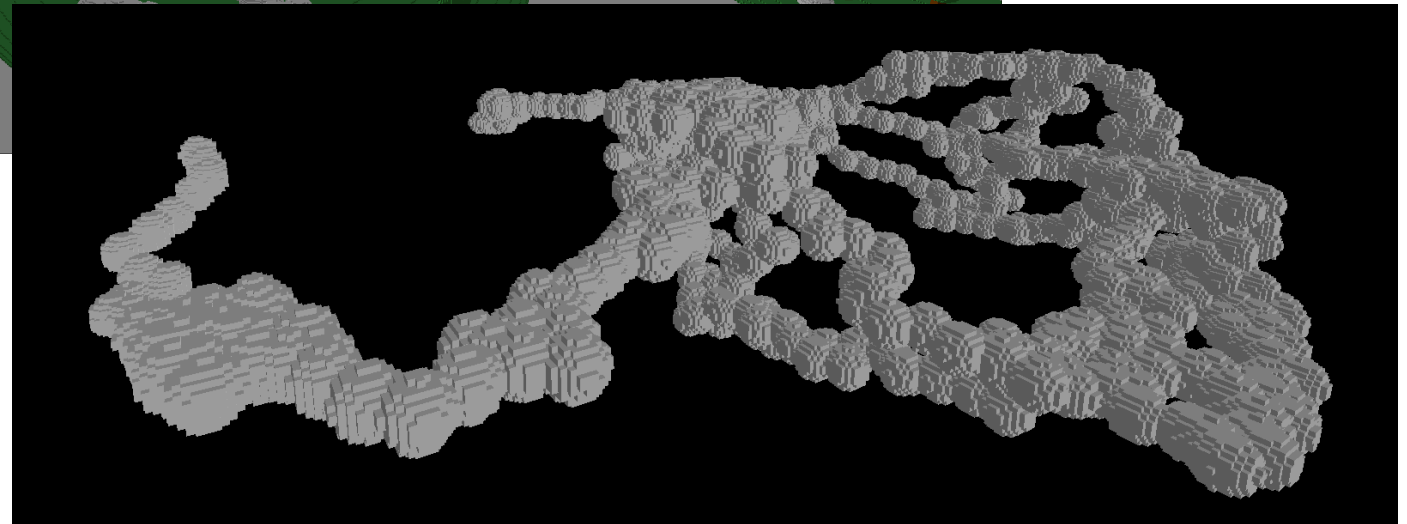


Analyse

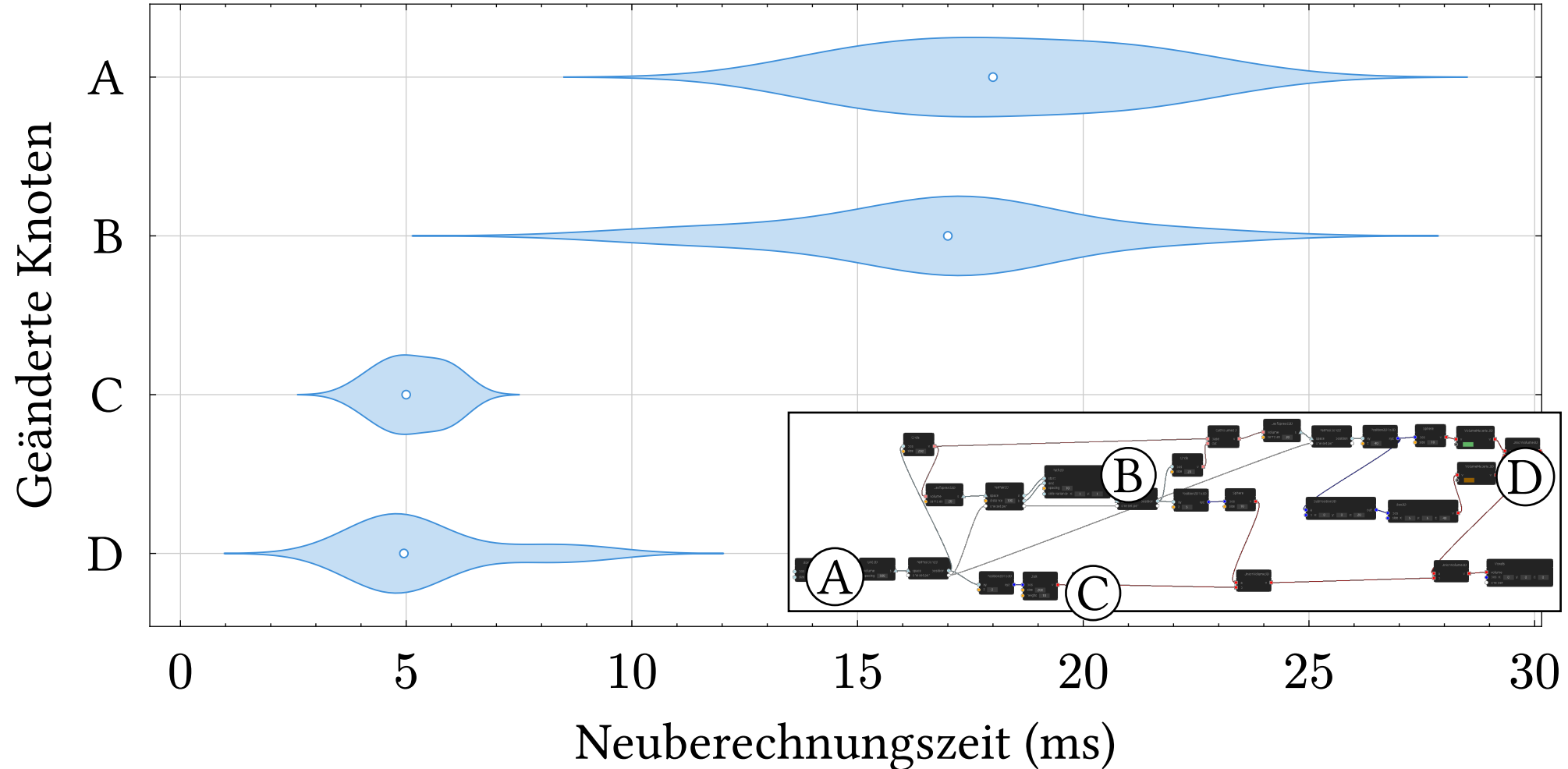


Insel-Beispiel

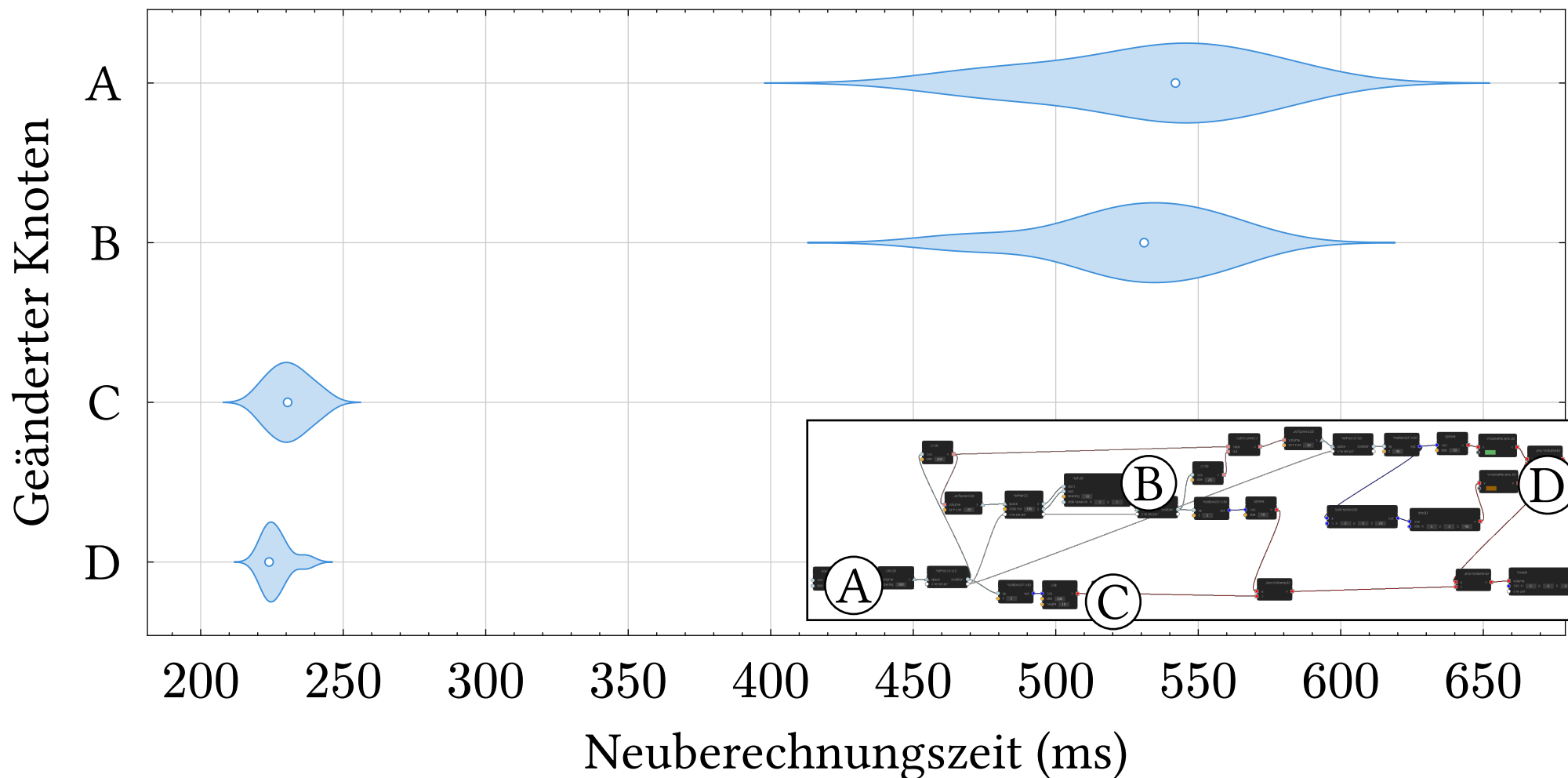
Höhlen-Beispiel



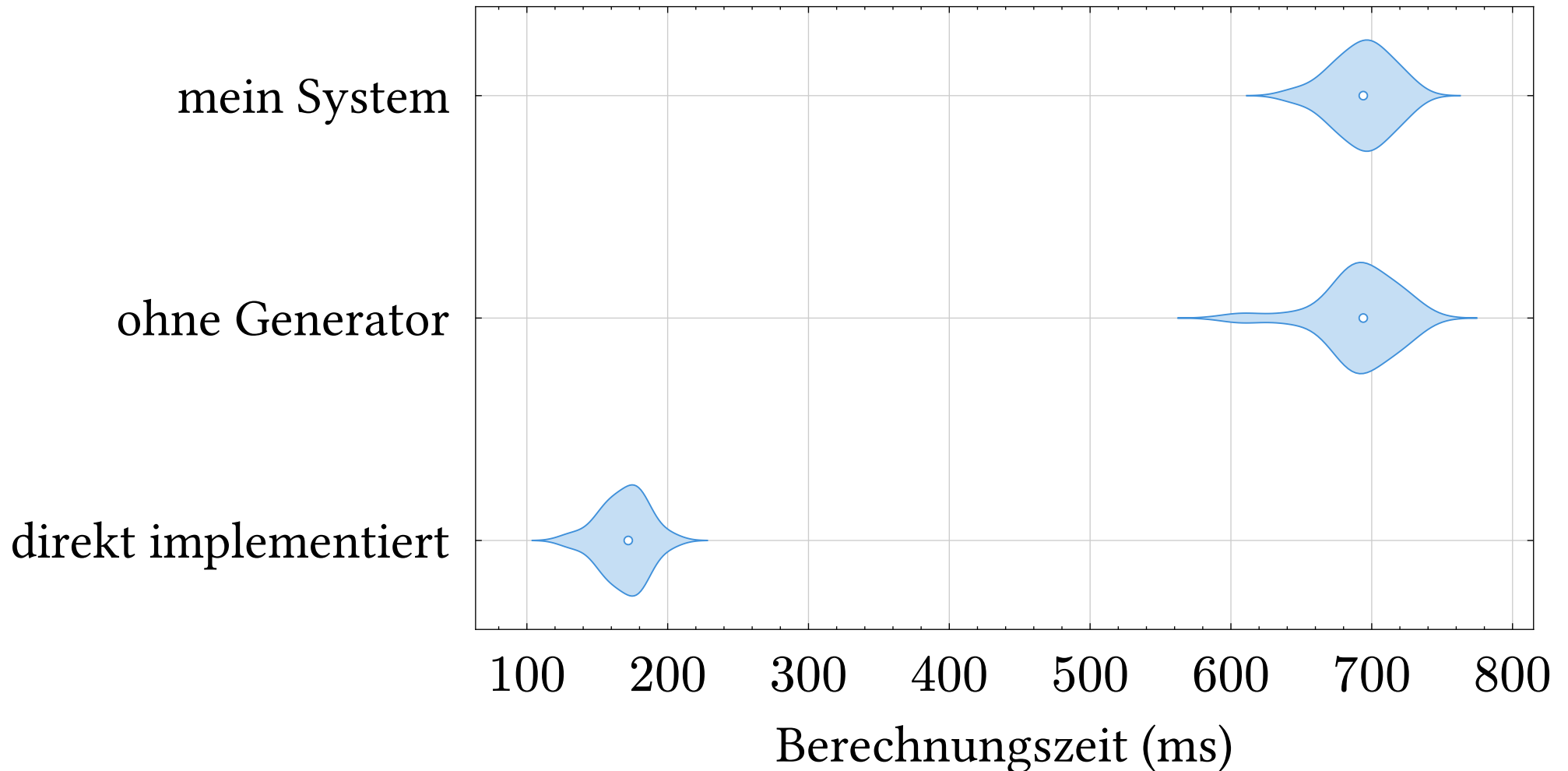
Insel-Beispiel (Generationsbereich: 2000^2 m)



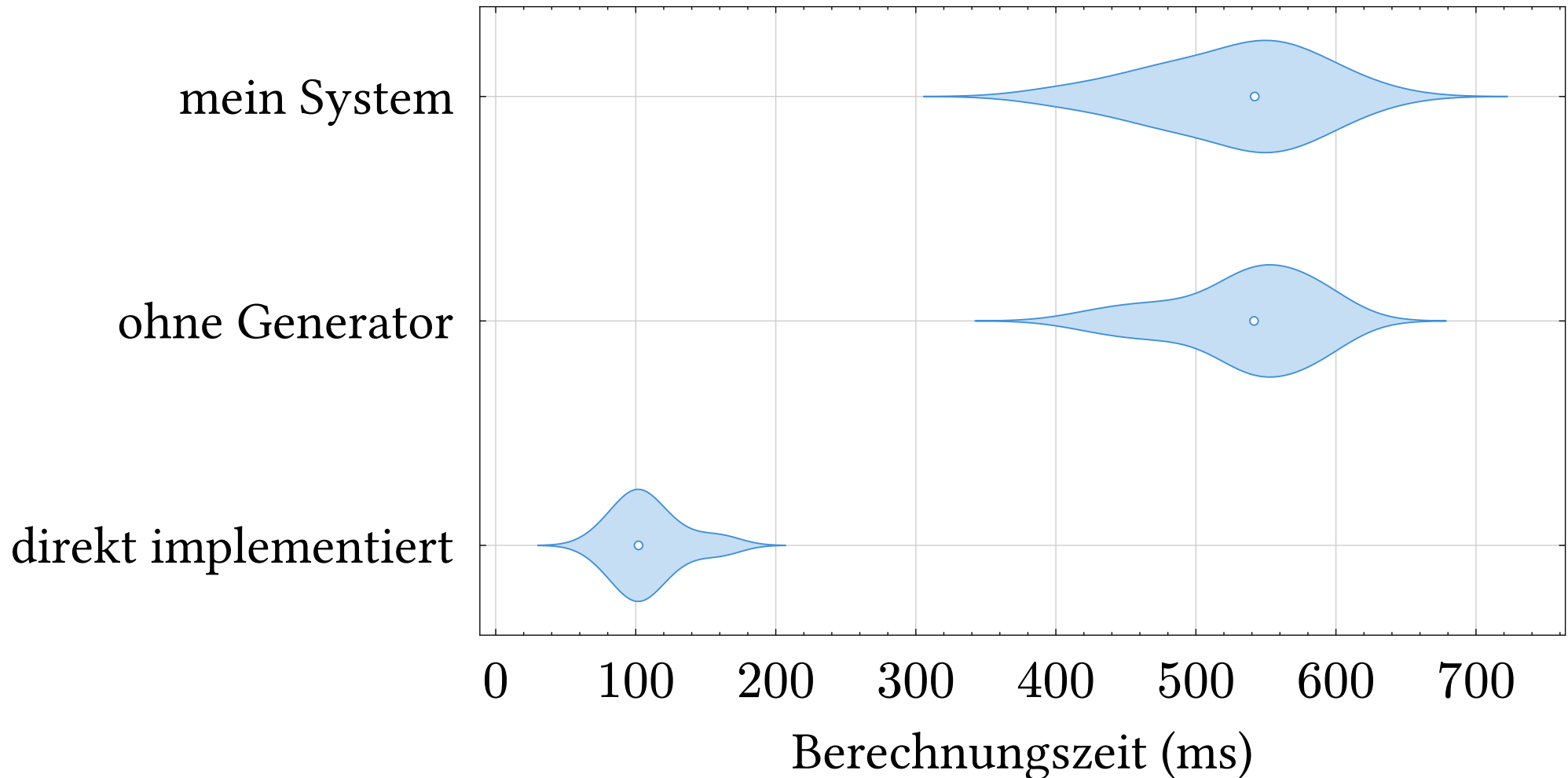
Insel-Beispiel (Generationsbereich: 20000^2 m)



Höhlen-Beispiel



Insel-Beispiel



- Skalierung von sehr große Welten
- Integration in bestehende Game Engines
- Parallelisierung der Generierung
- Automatische Cache-Optimierung
- Kreis-Abhängigkeiten & Leere Lösungen
- Experten-Meinungen zur Bewertung der Nützlichkeit

Beispiel Videos

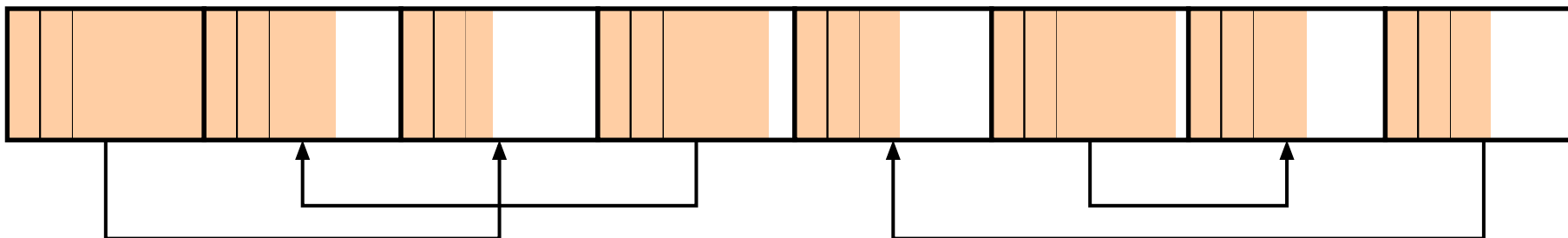
Vielen Dank.

Fragen?

```
enum TemplateValue {  
    Number(NumberValue),  
    Position(PositionValue),  
    Volume(VolumeValue),  
    ...  
}
```

```
enum PositionValue {  
    Add((PosIndex, PosIndex)),  
    Sub((PosIndex, PosIndex)),  
    Numbers([NumberIndex; 3]),  
    ...  
}
```

```
let values: Vec<TemplateValue>
```



```
fn make_position(child_node: &EditorNode,  
                 in_index: usize) -> PosIndex {
```

```
    let node = child_node.inputs[in_index];
```

```
    let value = match &node.data_type {
```

```
        EditorNodeType::Add => {
```

```
            let a = make_position(node, 0, ...);
```

```
            let b = make_position(node, 1, ...);
```

```
            TemplateValue::Position(PositionValue::Add((a, b)))
```

```
        },
```

```
        EditorNodeType::Sub => ...
```

```
    }
```

```
fn calc_position(index: PosIndex) -> SmallVec<Vec3> {  
  
    match values[index] {  
        PositionValue::Add((a_index, b_index)) => {  
            let a = calc_position(a_index);  
            let b = calc_position(b_index);  
  
            return a.cartesian_product(b)  
                .map(|(a_v, b_v)| a_v + b_v);  
        },  
        PositionValue::Sub => ...  
    }  
}
```

- **Minecraft:** eigener Screenshot
- **Layered Noise:** <https://velog.velcdn.com/images/suhan0304/post/5decd2a9-bbfa-43fc-9cb1-b17b6c4944e5/image.png>
- **Blender:** <https://www.reddit.com/media?url=https%3A%2F%2Fpreview.redd.it%2Fneed-some-clouds-geometry-nodes-v0-vt96ocz4z7w91.jpg%3Fauto%3Dwebp%26s%3D5eb3063e5dfc3694d7b99f80a96bdbfc42583c8c>
- **Houdini:** <https://i.pinimg.com/originals/45/8c/29/458c298b4ee094cf161f8e32f88d5505.jpg>
- **Tarrain diffusion:** <https://github.com/xandergos/terrain-diffusion>

Zugegriffen am: 7.9.2026 19 Uhr